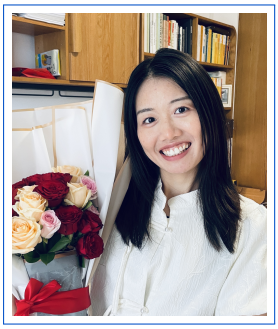
Fang Yang



Curriculum Vitae

Personal Information

Adresse	Vivatsgasse 7, MPIM, Bonn, 53111, Germany
E-mail	f.yang@mpim-bonn.mpg.de
Birthday	May 11, 1996
Homepage	https://fangyang-math.github.io/

Education

2014-2019	**Bachelor of Science**, *Sun Yat-Sen University*
2019-2024	**Doctor of Philosophy**, *Tsinghua University*, Supervisor: Fan Xu
2024-now	**Postdoc**, *Max-Planck Institute for Mathematics*, Bonn

Scholarships

2015-2019	Endeavor Scholarship (Awarded by Chinese Ministry of Eduation)
2018-2019	Outstanding graduate of Sun Yat-Sen University
2020-2023	Graduate Scholarship (Awared by Tsinghua University)

Research visits

2023.2-2023.6	**Visiting Ph.D. Student**, *Université Paris-Saclay*, under guidance of Professor Olivier Schiffmann

Interests

Algebraic Representation Theory	Calabi-Yau category and tilting theory, coherent sheaves on weighted projective lines, preprojective algebra and Hall algebra.
Geometric Representation Theory	Categorification, semicanonical basis and Nakajima quiver variety, cohomological Hall algebra
Combinatorial Theory	Quantum cluster algebra and Donaldson-Thomas invariants.

Publications

1. **Applications of spherical twist functors to Lie algebras associated to root categories of preprojective algebras**, *with Fan Xu*, Journal of Algebra, Volume 622 (2023): 556-586
2. **Motivic Cluster Multiplication Formulas in 2-Calabi-Yau Categories**, *with Jie Xiao and Fan Xu*, Int. Math. Res. Notices (2024): 13785-807

Preprints

1. **Quantum cluster algebras associated to weighted projective lines**, *with Fan Xu*, arXiv: 2207.02837, submitted
2. **KLR-Schur algebra of coherent sheaves on the projective line: Tilting and PBW bases**, *with Olivier Schiffmann*, arXiv: 2603.01265, submitted
3. **Degenerations in graded quiver varieties and in derived categories of Dynkin quivers**, *with Alessandro Contu*, arXiv: 2602.23939
4. **Formulas in quantum Grothendick ring via derived Hall algebras of type A**, *with Alessandro Contu*, In Preparation

Presentations

Cluster algebras

June 2023 A talk on *quantum cluster algebras of weighted projective lines* at the conference "Representation Theory in Lyon".

Sep. 2023 A talk on *quantum cluster algebras of weighted projective lines* at Beijing Normal University.

Nov. 2023 A talk on *quantum cluster algebras of weighted projective lines* at the conference "The 16th National Algebra Academic Conference".

Oct. 2024 A talk on *multiplication formulas in 2-Calabi-Yau categories* at University of Cologne.

2025.03 A talk on *quantum cluster algebras of weighted projective lines* at the conference "Workshop on bricks and endofinite representations".

CoHA

Oct. 2025 A talk on *KLR-Schur algebra of the projective line* at the conference "Collaborations in Algebra, Representation theory and Ethics".

Teaching

2019-2020 Fall **Teaching Assistant in Abstract Algebra I**

Each semester in 2020-2022 **Teaching Assistant in Linear Algebra**

Else

Conferences

Jul. 2023 **Geometric Representation Theory and Applications**, *Tsinghua University*

Jul. 2023 **International Congress of Basic Science**, *Beijing City government*

Nov. 2023 **Algebraic Representation Theory**, *Beijing Normal University*

Aug. 2024 **International Conference on Representations of Algebras (ICRA 21, 2024)**, *Shanghai Jiao Tong University*

Mar. 2025 **Algebra Days in Caen 2025: Categorization and quantification in cluster theory and beyond**, *University of Caen-Normandy*

Apr. 2025 **Enveloping Algebras and Geometric Representation Theory**, *Mathematisches Forschungsinstitut Oberwolfach*

Jun. 2025 **Enumerative Geometry and Geometric Representation Theory**, *Università di Pisa*

Sep. 2025 **ARTA X: Advances in Representation Theory of Algebras**, *University of Cologne*

Seminars

Spring 2022 **Deriving DG categories**, *Bernhard Keller*,
Reference: Deriving DG categories, Bernhard Keller.

Fall 2022 **Cluster algebras and scattering diagrams**, *Yutong Yu*,
Reference: Wall-Crossing Structures in Cluster Algebras, Lang Mou.

Fall 2022 **Quiver varieties and Coulomb branches**, *Dylan G Allegretti, Peng Shan*,
Reference: K-theoretic Coulomb Branches of quiver Gauge theories and cluster varieties, Gus Schrader, Alexander Shapiro.

Fall 2023 **Integral homology of loop groups and affine Grassmannians**, *Boming Jia, Peng Shan*,
Reference: An introduction to affine Grassmannians and the geometric Satake equivalence, Xinwen Zhu.